

Refining Treatment Monitoring in Patients With Relapsing Multiple Sclerosis Based on Changes on Neurofilament Light Chain Z-Scores

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Neurology® 2026;107:e218451. doi:10.1212/WNL.00000000000218451

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Abstract

Background and Objectives

Serum neurofilament light chain (sNfL) reflects inflammatory axonal damage in relapsing multiple sclerosis (RMS). Longitudinal sNfL dynamics may aid treatment response assessment, although data on expected changes under disease-modifying treatments remain limited, particularly in real-world practice. This study evaluated whether on-treatment changes in sNfL were associated with subsequent evidence of disease activity (EDA) and whether they provided complementary information to established treatment response scoring systems (TRSS).

Methods

This prospective, real-world cohort study was conducted at the Multiple Sclerosis Center of Catalonia (Cemcat). Adult RMS patients initiating disease-modifying treatments (DMTs) with serum samples at treatment initiation (baseline) and after 1 year (Y1) were included. The primary exposure was change in sNfL z-scores ($\Delta zNfL = zNfL_{Y1} - zNfL_{baseline}$) analyzed using data-derived thresholds ($\Delta zNfL \geq 0$ and <0.5 -point reduction). At Y1, Expanded Disability Status Scale progression, clinical/radiologic activity, and TRSS (Rio [RS], modified Rio [mRS], and MAGNIMS) were assessed. The primary outcome was EDA between years 1 and 2 (EDA-Y2). Multivariable logistic regression models evaluated associations between $\Delta zNfL$ and EDA-Y2, as well as the added value of $\Delta zNfL$ across TRSS strata.

Results

A total of 329 patients (66.3% female; median age 39 years) initiated 352 DMTs; 342 remained on treatment for ≥ 1 year and 182 had complete EDA-Y2 data. Mean (SD) $\Delta zNfL$ at Y1 was -0.62 (1.2), with greater reductions in monoclonal antibody-treated and treatment-naïve patients. Lack of zNfL reduction over the first year ($\Delta zNfL \geq 0$) was associated with higher odds of EDA-Y2 in the overall cohort (OR [95% CI], p : 3.07 [1.47 – 6.39], 0.003), in treatment-naïve cases (14.04 [2.47 – 79.93], 0.003), and in those achieving No Evidence of Disease Activity (NEDA-3) at Y1 (4.54 [1.26 – 16.30], 0.02). Similar associations were observed for <0.5 -point reductions. Adding $\Delta zNfL$ to TRSS showed persistent associations with EDA-Y2 despite otherwise favorable scores [e.g., RS: 2.26 (1.12 – 4.59), 0.024].

Discussion

Failure to reduce zNfL levels after 1 year of treatment was associated with subsequent disease activity, even in patients otherwise considered stable, providing complementary information to clinical and MRI-based monitoring. Limitations include incomplete follow-up and the modest discriminative performance of $\Delta zNfL$, supporting a complementary rather than stand-alone role for zNfL dynamics.

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Glossary

Δ zNfL = changes in neurofilament light chain z-scores at year 1; BMI = body mass index; DMT = disease-modifying treatment; EDA = evidence of disease activity; EDSS = Expanded Disability Status Scale; mRS = modified Rio; MS = multiple sclerosis; NEDA = no evidence of disease activity; NfL = neurofilament light chain; OR = odds ratio; RMS = relapsing multiple sclerosis; RR = rate ratio; RS = Rio; sNfL = serum neurofilament light chain; TRSS = treatment response scoring systems; Y1 = year 1; Y2 = year 2.

Introduction

Reliable biomarkers that capture inflammatory activity and treatment response are needed in multiple sclerosis (MS) to support clinical decision making. Neurofilament light chain (NfL), a biomarker of neuroaxonal damage, is elevated in CSF and blood in MS^{1,2} and has been consistently associated with relapses and focal inflammatory lesions on brain MRI. These properties make NfL a dynamic biomarker of inflammatory disease activity.³⁻⁷

Advances in highly sensitive detection methods now allow for precise quantification of serum NfL levels (sNfL), enabling longitudinal monitoring in routine clinical practice.⁸ To facilitate clinical interpretation, NfL concentrations can be expressed as z-scores, representing the number of standard deviations from age-adjusted and body mass index (BMI)-adjusted population means, where 0 indicates expected levels and higher values reflect higher-than-expected concentrations. This approach allows comparison with expected normative values and supports longitudinal intraindividual analyses, overcoming limitations of absolute concentrations.⁹

Elevated sNfL also serve as prognostic indicators for future disease activity and related disability accumulation.⁹⁻¹¹ sNfL decrease after initiation of disease-modifying treatments (DMTs) and associate with drug effect and have therefore been incorporated into clinical trials as surrogates of treatment response. In these populations, persistently high sNfL despite treatment may reflect suboptimal control of disease activity.^{4,5,12-14}

This association holds even in patients receiving high-efficacy therapies, in whom elevated sNfL may signal ongoing damage despite suppressed focal inflammation.¹⁵ Furthermore, patients with no evidence of disease activity (NEDA), but elevated sNfL seem to be at increased risk of subsequent evidence of disease activity (EDA).⁹

Despite its potential, integrating sNfL into routine treatment monitoring remains challenging because of the lack of consensus on the magnitude of post-treatment changes that should be expected under effective therapy. Questions remain about whether sNfL z-scores (zNfL) or concentrations should normalize or drop below a specific threshold. Another pertinent question is whether incorporating sNfL data enhances existing scoring systems for measuring treatment response. In addition, the applicability of sNfL thresholds across different DMTs and patient populations requires further clarification.

This study aims to evaluate the association between 1-year changes in zNfL (Δ zNfL) after DMT initiation and subsequent EDA in a real-world cohort of patients with relapsing MS (RMS). We further assess the added value of Δ zNfL when combined with established treatment response scoring systems (TRSS). Finally, we explore these associations in clinically relevant subgroups, including treatment-naïve patients and those achieving NEDA-3 status at year 1.

Methods

Study Design

This single-center cohort study included adult patients with relapsing multiple sclerosis (RMS) who initiated a disease-modifying therapy (DMT) between March 2003 and December 2021, with follow-up data available through February 2023. Both treatment-naïve and previously treated patients were included (Figure 1). All patients fulfilled the 2017 revised McDonald diagnostic criteria.

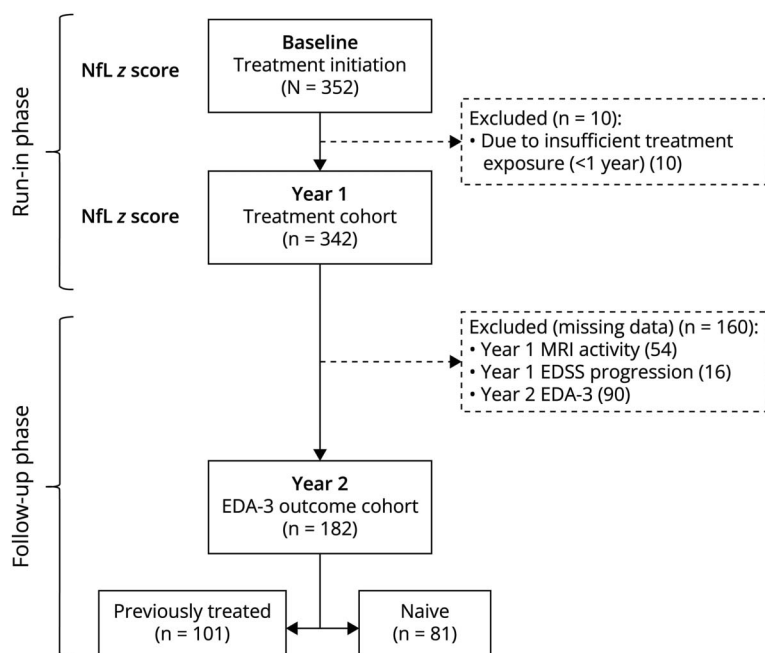
Eligible patients were required to have serum samples collected at treatment initiation (baseline, ± 90 days) and after 1 year of continuous treatment with the same DMT (Y1, ± 90 days). Patients who discontinued or switched therapy during the first year were excluded from longitudinal analyses. When patients changed treatment after the first year, only data corresponding to the original DMT were considered.

During the 12-month run-in phase, relapses, Expanded Disability Status Scale (EDSS) progression, and MRI activity were prospectively recorded.

EDSS progression was defined as an increase of ≥ 1.5 , 1.0, or 0.5 points according to baseline EDSS (<1.0 , 1–5.5, or >5.5 , respectively), assessed during clinically stable periods, ≥ 30 days after relapse resolution, and in the absence of corticosteroid treatment, consistent with standard assessment practice. MRI activity was defined as ≥ 1 neT2 and/or ≥ 1 CEL.

Brain MRI scans were acquired on 1.5 or 3.0 T scanners following a standardized protocol.¹⁶ Scans performed within ± 180 days from baseline [median (IQR): -69.5 (-113 – (-23))] and Y1 [-2 (-20 to 15)] were included. The interval between MRI acquisition and serum sampling was shorter, with median differences of -66.5 days (-113 to -16.3) at baseline and -15 days (-36 to 15) at year 1. MRI activity was assessed visually by an experienced rater.

Figure 1 Study Design and Case Selection



During the run-in phase, 352 treatment initiations were recorded at baseline. Ten cases were excluded because of insufficient treatment exposure (<1 year), yielding a year 1 treatment cohort of 342 cases. For the year 2 outcome analyses, cases were excluded because of missing year 1 MRI activity (n = 54), missing year 1 EDSS progression data (n = 16), or unavailable EDA-3 status at year 2 (n = 90), resulting in a final Y2 EDA (outcome) cohort of 182 cases. Among them, 101 were previously treated and 81 were treatment-naïve. EDA = evidence of disease activity; EDSS = Expanded Disability Status Scale; NfL = neurofilament light chain; Y1 = year 1; Y2 = year 2.

Serum samples obtained at baseline and Y1 were aliquoted and stored at -80°C following standardized procedures.¹⁷ The median interval between baseline and follow-up sampling was 12.5 months (IQR 11.96–13.17). sNfL levels were measured in duplicate using single molecule array technology (Simoa, Quanterix, Neurology 2-plexB assay). Age-adjusted zNfL values were calculated based on external healthy controls.⁹ Because BMI data were not systematically available, the default value of 25 specified by the z-score calculator was applied, in accordance with the tool's specifications. Changes in serum zNfL were quantified as $\Delta\text{zNfL} = \text{zNfLY1} - \text{zNfL baseline}$.

At Y1, TRSS including Rio (RS), modified Rio (mRS), and MAGNIMS scores were calculated and dichotomized into low (0–1) or high (2–3 for RS/mRS, 2 for MAGNIMS) categories.^{18–20}

During the follow-up phase, the outcome of interest—evidence of disease activity at year 2 (EDA-Y2)—was defined as the occurrence of any relapse, confirmed EDSS progression, or MRI activity between year 1 and year 2 after DMT initiation.

Statistical Analysis

Baseline demographic, clinical, and radiologic characteristics were compared between subgroups using χ^2 tests for categorical variables and *t* tests or Mann–Whitney *U* tests for continuous variables depending on distribution. Variables showing between-group differences were included as covariates in sensitivity adjusted models.

Associations between baseline and Y1 zNfL and relapses, neT2 lesions, and CEL counts were analyzed using Poisson regression reporting rate ratios (RR). EDSS, treated as an ordinal variable, was analyzed using ordinal logistic regression; when sparse categories violated proportional-odds assumptions, EDSS was grouped into clinically meaningful strata (0–1.5; 2.0–3.5; 4.0–5.5; ≥ 6.0).

Paired comparisons (paired *t* tests) were used to describe within-patient changes in zNfL between baseline and Y1, and ANOVA was used to compare mean ΔzNfL across DMT categories.

To define cut-points, ROC curves for ΔzNfL and Y1 zNfL were generated. For each coordinate, the Youden index ($J = \text{sensitivity} + \text{specificity} - 1$) was calculated. ΔzNfL was the primary longitudinal exposure of interest, reflecting treatment-associated change between baseline and year 1. Y1 zNfL was evaluated as a secondary cross-sectional on-treatment biomarker.

For Y1 zNfL, the maximal *J* occurred around 1.4–1.6, supporting $\text{zNfL} \geq 1.5$ as an operational cut-point. For ΔzNfL , *J* values formed a plateau between 0 and -0.5 , leading to 2 prespecified and clinically interpretable thresholds: $\Delta\text{zNfL} \geq 0$ (no reduction) and $\Delta\text{zNfL} > -0.5$ (suboptimal reduction). An additional exploratory threshold near the Youden optimum ($\Delta\text{zNfL} -0.3$) was examined in sensitivity analyses. These cut-points were used in univariable and multivariable logistic regression models evaluating associations with EDA-

Y2, adjusting for age, disease duration, clinical and MRI activity, and treatment category.

Composite models integrating Δ zNfL with RS, mRS, and MAGNIMS scores were constructed. Patients were categorized according to combinations of low/high TRSS and reduced/not reduced Δ zNfL to assess associations with the outcome EDA-Y2.

Because only a subset of the baseline cohort had complete information to determine EDA at year 2, we conducted predefined sensitivity analyses to explore and mitigate potential selection bias. Baseline variables that differed between groups were incorporated as covariates in additional adjusted logistic regression models. To further address potential attrition bias, we applied inverse probability-of-attrition weighting. A logistic model was used to estimate each participant's probability of being included in the analytic cohort, and the inverse of this probability was applied as a weight in the association models evaluating Δ zNfL and EDA-Y2.

Analyses were repeated in treatment-naïve patients, previously treated and in patients achieving NEDA-3 status at Y1. To assess stability of estimates in smaller subgroups, non-parametric bootstrap logistic regression (1,000 replications) was performed using the same model specifications.

All statistical analyses were performed using SPSS v26.0. A 2-sided p value <0.05 was considered statistically significant. This study was reported in accordance with the STROBE reporting guideline.

Standard Protocol Approvals, Registrations, and Patient Consents

The study protocol was approved by the ethics committee of Vall d'Hebron University Hospital, and written informed consent was obtained from all participants.

Data Availability

The data that support the findings of this study are available from the corresponding author on reasonable request.

Results

Cohort Baseline Characteristics

This cohort study included 329 patients with RMS (218 female patients, 66.3%) who initiated 352 DMT courses, with a median follow-up of 7.6 years (IQR 3.6–14.6). The median age at treatment initiation was 39 years (IQR 31–45), and 198/352 cases (56.2%) were treatment-naïve. Median disease duration at baseline was 5.3 years (IQR 1.32–12.08).

Platform-injectable therapies (interferon and glatiramer acetate) accounted for 153 treatment initiations; oral therapies (teriflunomide, dimethyl fumarate, fingolimod, and cladribine) for 89; and monoclonal antibodies (rituximab,

ocrelizumab, alemtuzumab, and natalizumab) for 110. At baseline, 289/352 (82.1%) had clinical inflammatory activity in the previous 2 years and 132/265 (49.8%) ≥ 1 contrast-enhancing lesion on baseline MRI.

Of the 352 cases, 182 had complete information to determine evidence of disease activity at year 2 (EDA-Y2) and were included in the primary regression analyses. Baseline characteristics of included vs excluded cases are shown in Table 1, demonstrating no significant differences in sex, baseline clinical activity, baseline MRI activity, or baseline zNfL levels. Differences were observed in prior treatment exposure, baseline relapse count, and baseline CEL count, which were accounted for in sensitivity analyses. Among treatment-naïve patients, 81/198 had complete EDA-Y2 data and were included in subgroup analyses.

For TRSS calculations, RS was available in 179 cases, mRS in 183, and MAGNIMS in 192. Baseline characteristics are summarized in Table 1.

NfL Z-Scores Associate With Inflammation at Baseline and Y1

Poisson regression models showed a positive association between baseline zNfL and the number of relapses in the previous 2 years (RR [95% CI], p : 1.19 [1.10–1.28], <0.001 , $n = 352$) and number of CELs in brain MRI (2.41 [2.19–2.66], <0.001 , $n = 265$). Mean (SD) zNfL were higher in 158 cases with MRI activity vs 59 without [1.93 (1.11) vs 0.80 (1.08), $p < 0.001$]. Ordinal logistic regression using categorized EDSS at baseline, showed no significant association with baseline zNfL (OR [95% CI], p : 0.96 [0.81–1.13], 0.597; $n = 350$).

At Y1, zNfL remained associated with number of relapses (1.56 [1.27–1.92], <0.001 , $n = 342$), neT2 lesions (1.97 [1.81–2.14], <0.001 , $n = 281$), and CELs (4.09 [3.37–4.95], <0.001 , $n = 262$), the latter corresponding to an approximately fourfold increase in lesion rate per unit increase in zNfL. In addition, mean [SD] zNfL in 127 cases with MRI activity during Y1 were higher compared with 161 without (1.24 [1.05] vs 0.70 (0.94), $p < 0.001$). Categorized EDSS at Y1 showed a modest association with zNfL (1.33 [1.01–1.77], 0.045, $n = 309$).

The results were similar for the group of treatment-naïve patients (Table 2) and remained consistent after adjusting for age, disease duration, and previous treatment (eTable 1).

Changes in zNfL After DMT Initiation

Mean zNfL significantly decreased from baseline to Y1 in the overall cohort (1.55 [SD 1.22] to 0.93 [1.06]; $p < 0.001$), corresponding to a mean Δ zNfL of -0.62 [1.22], or an approximate 40% reduction.

Reductions differed by treatment category, being smallest under oral therapies (mean Δ zNfL -0.40 [1.18]), intermediate with platform injectables (-0.56 [1.27]), and greatest with monoclonal antibodies (-0.87 [1.13]). Reductions in zNfL were

Table 1 Baseline Characteristics of the Study Cohort

	Baseline cohort N = 352	Y1 cohort N = 342	Y2 (EDA) cohort N = 182	Y2 excluded cases N = 160	Y2 included vs excluded p Value
Age—median (IQR)	39 (31–45)	39 (32–45)	38.5 (31–44)	40 (32–45.75)	0.366
Sex F/M—(F %)	237/115 (67.3%)	229/113 (66.9%)	124/58 (68.1%)	105/55 (65.6%)	0.623
Disease duration at baseline (Y)—median (IQR)	5.32 (1.32–12.08)	5.31 (1.35–12.13)	5.75 (1.48–12.03)	4.789 (1.28–12.44)	0.628
Time of follow up 2023 from baseline (Y)—median (IQR)	7.56 (3.65–14.62)	7.72 (3.65–14.71)	5.86 (3.42–9.21)	10.46 (4.45–16.02)	<0.001
Treatment-naïve, n (%)	198 (56.2)	192 (56.1)	81 (44.5)	111 (69.4)	< 0.001
Previously treated (anytime), n (%)	154 (43.8)	150 (43.9)	101 (55.5)	49 (30.6)	< 0.001
Baseline clinical activity (YES), n (%)	289 (82.1)	150 (71.8)	154 (84.6)	129 (80.6)	0.330
Time between last relapse/BL sampling (mo)	5.06 (2.81–8.54)	5.32 (2.76–8.84)	4.5 (2.48–8.20)	5.5 (3.10–9.08)	0.108
Relapses 2 y prior initiation—median (IQR)	1(1–2)	1 (1–2)	1 (1–2)	1 (1–2)	0.014
Baseline EDSS—median (IQR)	2 (1.5–3.0)	2 (1.5–3.0)	2 (1.5–3.5)	2 (1–2.5)	0.064
Baseline MRI activity, n (%)					0.821
YES	158 (44.9)	150 (43.9)	89 (49)	61 (38)	
NO	59 (16.8)	59 (17.3)	34 (19)	25 (16)	
NA	135 (38.4)	133 (38.8)	59 (32.4)	74 (46.3)	
Baseline CEL (YES)—median (IQR)	0 (0–2) n = 265	0 (0–2) n = 272	1 (0–2) n = 143	0 (0–1) n = 114	0.043
NfL z-score—mean (SD)	1.56 (1.21)	1.55 (1.22)	1.65 (1.15)	1.44 (1.28)	0.106

Abbreviations: BL = baseline; CEL = contrast-enhancing lesion; DMT = disease modifying treatment; EDA = evidence of disease activity; EDSS = Expanded Disability Status Scale; F = female; M = male; mo = months; NfL = neurofilament light chain; Y = years; IQR = interquartile range; SD = standard deviation; NA = not available (baseline MRI activity could not be assessed because of lack of previous MRI and/or no gadolinium administration). Baseline characteristics of the 352 treatment initiations at baseline, the 342 cases completing 1 year of continuous therapy, and the 182 cases included in the primary association analyses for EDA-Y2, compared with the 160 cases excluded because of incomplete year-2 follow-up data. Variables are shown to allow assessment of potential selection bias.

Note: Baseline corresponds to the time of DMT initiation. De-escalation was considered when a treatment switch was made from mAbs to oral/platform-injectable DMT or from oral to platform-injectables.

DMTs: (1) Interferon B1a (n = 65), Interferon B1b (n = 50), Glatiramer acetate (n = 38); (2) Teriflunomide (n = 20), Dimethyl fumarate (n = 28), Cladribine (n = 9), Fingolimod (n = 32); (3) Alemtuzumab (n = 12), Rituximab (n = 17), Ocrelizumab (n = 33), Natalizumab (n = 48).

greater in patients treated with monoclonal antibodies than in those receiving oral therapies ($p = 0.024$).

In treatment-naïve patients, changes were more pronounced, particularly in the mAb subgroup (mean $\Delta zNfL$ -1.67 [1.16]) (Figure 2).

Association Between $zNfL$ and EDA

$\Delta zNfL$ Is An Independent Biomarker Associated With EDA-Y2

The association between $\Delta zNfL$ and EDA-Y2 was evaluated in the overall cohort, followed by predefined subgroup, sensitivity, and complementary analyses.

Absence of $zNfL$ reduction at Y1 ($\Delta zNfL \geq 0$) was associated with higher odds of EDA-Y2 in multivariable models, adjusted for inflammatory activity (relapses, MRI activity) and EDSS progression at Y1 (OR [95% CI], p) (3.07 [1.47 – 6.38], 0.003 ;

$n = 182$), and remained after adjustment for treatment category (3.24 [1.41 – 7.42], 0.006).

Similarly, not reaching a ≥ 0.5 -point reduction in $zNfL$ was associated with higher odds of EDA-Y2 (2.33 [1.19 – 4.55], 0.014). After adjusting for treatment category, the association was attenuated (2.13 [0.99 – 4.60], 0.053).

Sensitivity and weighted analyses yielded effect estimates similar in magnitude and direction to the primary models, indicating that the associations were robust to adjustment for baseline imbalances and attrition (eTable 2.1).

Consistent results were observed across predefined subgroups. In treatment-naïve patients with complete EDA-Y2 data ($81/198$), lack of $zNfL$ reduction was strongly associated with EDA-Y2 (14.04 [2.47 – 79.93], 0.003), remaining significant after treatment adjustment (17.88 [1.71 – 186.4], 0.016). A ≥ 0.5 -point

Table 2 Associations Between Baseline and Year-1 zNfL and Biomarkers of Disease Activity

	#Relapses		EDSS		#CEL		#neT2	
	Baseline	Y1	Baseline	Y1	Baseline	Y1	Baseline	Y1
A. All cases								
RR/OR	1.19	1.56	0.96	1.33	2.41	4.09	—	1.97
CI 95%	1.10–1.28	1.27–1.92	0.81–1.13	1.01–1.77	2.19–2.66	3.37–4.95		1.81–2.14
p value	<0.001	<0.001	0.597	0.045	<0.001	<0.001	—	<0.001
N =	352	342	350	309	265	262	—	281
B. Treatment-naïve								
RR/OR	1.13	1.59	0.95	1.36	2.31	3.91	—	1.83
CI 95%	1.02–1.24	1.22–2.07	0.76–1.18	0.88–2.11	2.04–2.61	3.07–4.97		1.65–2.02
p value	0.015	<0.001	0.644	0.169	<0.001	<0.001	—	<0.001
N =	198	192	197	161	155	141	—	147

Abbreviations: CEL = contrast-enhancing lesion; EDSS = Expanded Disability Status Scale; neT2 = new-enlarged T2-weighted lesion; zNfL = neurofilament light chain z-score.
zNfL baseline and Y1 associations with count outcomes (relapses, contrast-enhancing lesions, and new/enlarged T2 lesions) were evaluated using Poisson regression models and are expressed as rate ratios (RR) with 95% confidence intervals. Associations with disability were assessed using ordinal logistic regression models for categorized EDSS and are expressed as odds ratios (OR) with 95% confidence intervals.

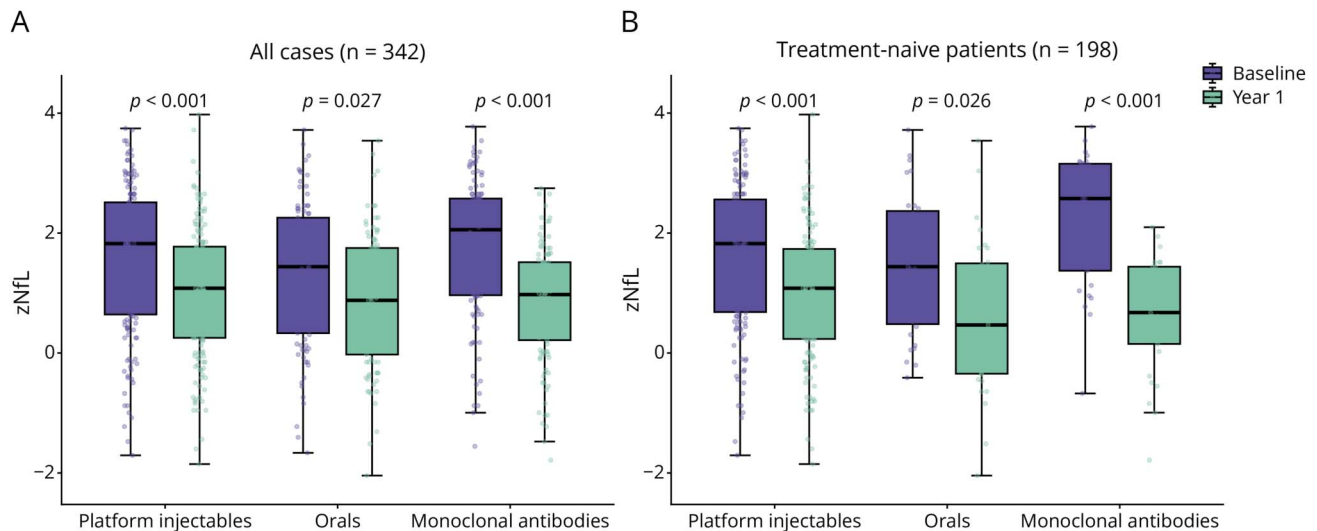
reduction threshold showed similar direction of effect (3.53 [1.15–10.78] 0.027), although significance was attenuated after adjustment (2.94 [0.69–12.62], 0.146). The results were also consistent in sensitivity analyses and inverse probability-of-attrition weighting (eTable 2.2).

In previously treated patients (n = 101), ΔzNfL ≥ 0 remained associated with EDA-Y2 (3.08 [1.15–8.28], 0.026), supporting the generalizability of this relationship.

Similarly, among patients achieving NEDA-3 at Y1 (n = 54), absence of zNfL reduction was associated with higher odds of EDA-Y2 (4.54 [1.26–16.3], 0.02), with comparable results observed using the ≥ 0.5-point reduction threshold (4.97 [1.33–18.55], 0.017) (Figure 3).

Bootstrap analyses performed across subgroups showed consistent direction and magnitude of effect, supporting the stability of these estimates (eTable 3).

Figure 2 Distribution of Serum Neurofilament Light Chain Z-Scores by Treatment Category



Box-and-scatter plots show the distribution of serum neurofilament light chain z-scores (zNfL) at baseline and at year 1 according to treatment category in the overall cohort (A) and in treatment-naïve patients (B). Boxes represent the median and interquartile range, with individual observations overlaid. p values correspond to paired t tests comparing mean zNfL at baseline and year 1. Mean changes (ΔzNfL) and corresponding standard deviations are reported in the Results.

Δ zNfL at Y1 Is Not Associated With EDA at Year 3

Exploratory analyses examined the association between Δ zNfL at Y1 and disease activity at year 3 (Y3). In univariable analyses, Δ zNfL at Y1 showed no association with EDA at Y3 overall (OR [95% CI], *p*) (0.999 [0.507–1.967], 0.997; *n* = 168), in treatment-naïve cases (2.24 [0.58–8.64], 0.241; *n* = 80), nor in those achieving NEDA-3 at Y1 (0.365 [0.100–1.334], 0.127; *n* = 51).

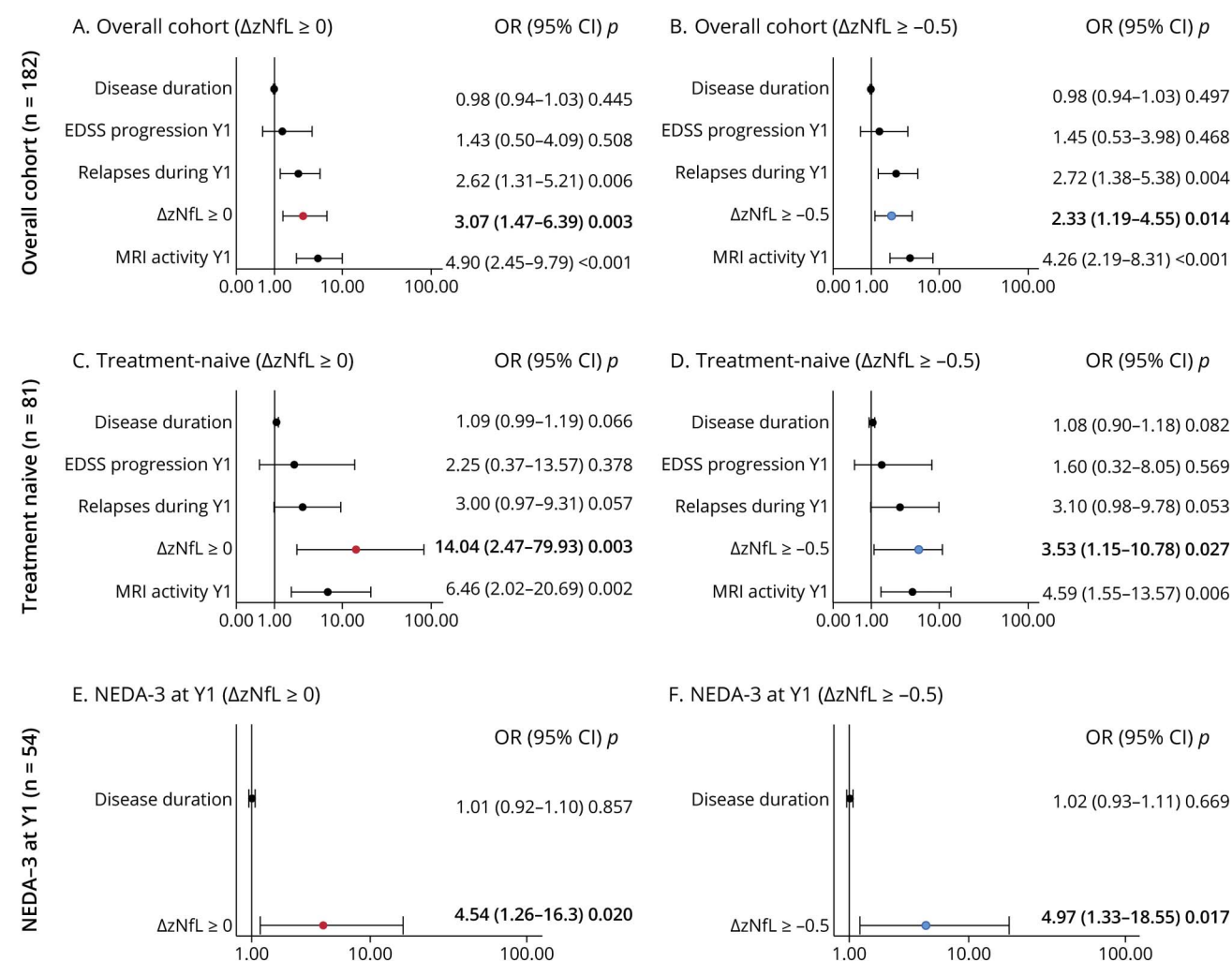
Similarly, multivariable models did not demonstrate an association between Δ zNfL and EDA-Y3 in the overall cohort (0.85 [0.41–1.77], 0.661; *n* = 152), in the naïve subgroup (1.71 [0.38–7.78], 0.487; *n* = 69), nor in patients meeting NEDA-3 at Y1 (0.41 [0.10–1.60], 0.199; *n* = 51).

Association Between a Single zNfL Measurement at Y1 and EDA

As a complementary cross-sectional analysis, the association between a single on-treatment zNfL measurement at Y1 and subsequent EDA was evaluated. Y1 zNfL ≥ 1.5 was associated with EDA-Y2 in univariable analyses (OR 2.88 [95% CI 1.59–5.19], *p* < 0.001).

In multivariable models adjusted for inflammatory activity at Y1 (*n* = 182), zNfL ≥ 1.5 remained associated with EDA-Y2 (2.10 [1.06–4.16], 0.033), and the association persisted after adjusting for treatment category (2.22 [1.02–4.84], 0.044). However, in patients with NEDA-3 at Y1 (*n* = 54), zNfL ≥ 1.5 was not associated with EDA-Y2 (2.25 [0.63–8.07], 0.214).

Figure 3 Association Between Δ zNfL and Subsequent Disease Activity at Year 2 (EDA-Y2)



Forest plots display multivariable logistic regression models assessing the association between changes in zNfL at year 1 (Δ zNfL) and evidence of disease activity at year 2 (EDA-Y2). Odds ratios (ORs) with 95% confidence intervals (CI) are shown. Panels A–B display results for the overall cohort: (A) Δ zNfL ≥ 0 (no reduction) and (B) Δ zNfL reduction < 0.5. Panels C–D show the corresponding analyses restricted to treatment-naïve patients and panels E–F to patients achieving NEDA-3 at year 1. Absolute number of cases/EDA-Y2 events included in each analysis: • Δ zNfL ≥ 0 : 182/86; • Δ zNfL reduction < 0.5: 182/86; • treatment-naïve subgroup: 81/48; • NEDA-3 at year 1 subgroup: 54/16. EDA = evidence of disease activity; EDSS = Expanded Disability Status Scale; NEDA = no evidence of disease activity; net2 = new/enlarged T2 lesion; Y1 = year 1; zNfL = neurofilament light chain z-score.

Y1 zNfL values were not associated with EDA-Y3 in univariable analyses, neither in the overall cohort (OR [95% CI], p : 1.54 [0.78–3.02], 0.210; n = 168), nor in treatment-naïve patients (1.26 [0.42–3.75], 0.682; n = 80), nor in those reaching NEDA-3 at Y1 (1.30 [0.36–4.69], 0.694; n = 51).

Consistently, multivariable models showed no significant associations overall (1.12 [0.54–2.34], 0.759; n = 152), in the naïve subgroup (1.01 [0.30–3.38], 0.981; n = 69), or in patients meeting NEDA-3 at Y1 (1.27 [0.35–4.65], 0.717; n = 51).

Added Value of Δ zNfL When Combined With Treatment Response Scoring Systems

The potential added value of Δ zNfL beyond established treatment response scores was also assessed. Across the cohort, high TRSS values were associated with substantially higher odds of EDA-Y2 vs low scores (RS OR 10.1 [3.37–30.23]; mRS OR 7.53 [2.11–26.84]; MAGNIMS OR 9.31 [2.64–32.81], all p < 0.01).

Among cases with low RS, absence of zNfL reduction was associated with higher odds of EDA-Y2 vs those with low RS and reduced zNfL (2.26 [1.12–4.59], 0.024). Similar findings were observed for mRS and MAGNIMS.

Cases with high TRSS and persistently high zNfL all experienced EDA-Y2, preventing OR estimation.

Failure to reach a 0.5-point reduction in Δ zNfL was similarly associated with higher odds of EDA-Y2 across low and high TRSS groups (e.g., RS low: 2.73 [1.37–5.45], 0.008; RS high: 42.05 [5.22–338.6], <0.001). The results were comparable for mRS and MAGNIMS and remained stable after adjusting for age, disease duration, and treatment category (eTables 4 and 5). The results are depicted in Figure 4.

Discussion

This study highlights the value of longitudinal changes in NfL after DMT initiation as a clinically informative biomarker of treatment response and disease monitoring in patients with RMS. Smaller reductions or persistently elevated zNfL levels after 1 year of treatment were associated with a higher likelihood of subsequent disease activity. By focusing on within-patient changes rather than single time-point measurements or population-based thresholds, this approach may help reduce the impact of interindividual and intraindividual variability and provide more clinically interpretable information for treatment monitoring in real-world clinical settings.^{2,17}

Consistent with prior evidence,^{4,21–23} zNfL showed a robust and reproducible association with inflammatory disease activity at both baseline and year 1, including clinical relapses and MRI markers of inflammation. These associations were observed across different analytical approaches and remained stable in treatment-naïve patients and after adjustment for

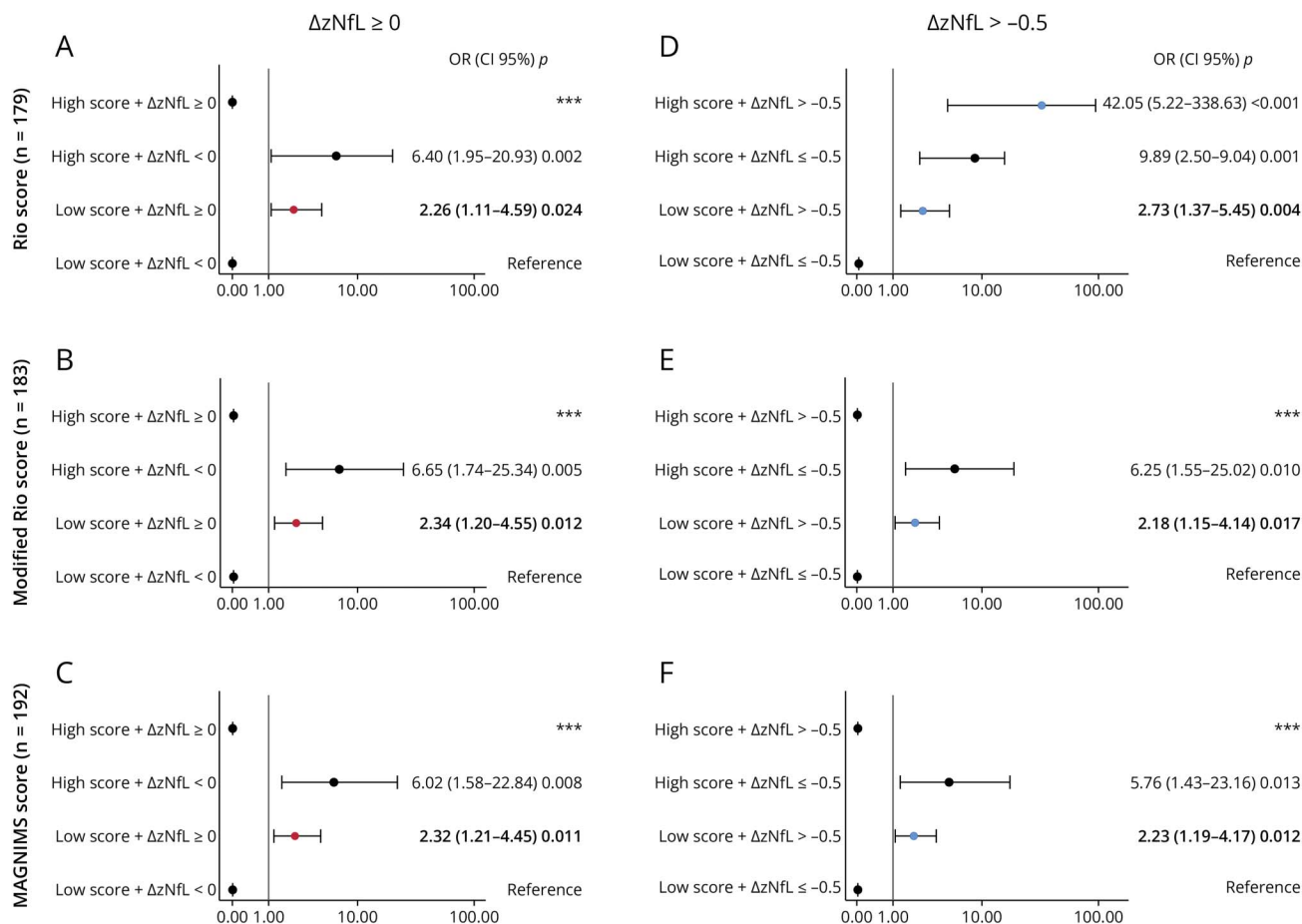
relevant clinical covariates. By contrast, no relevant association was observed between zNfL and EDSS when disability was assessed during clinically stable periods, supporting the concept that NfL primarily reflects ongoing inflammatory axonal injury rather than accumulated disability.

Previous studies have explored NfL alone or in combination with GFAP dynamics under treatment in relation to disability outcomes but were limited to progressive forms of MS or particular treatments and did not assess overall disease activity.²⁴ Other work has examined the prognostic value of a single NfL measurement at the time of the diagnosis for long-term risk of disability accumulation, accounting for treatment exposure, although the predictive capacity of NfL and the influence of treatment on its levels or temporal dynamics were not evaluated.²⁵ By contrast, our study evaluated zNfL dynamics in a heterogeneous, real-world RMS cohort initiating a broad range of DMTs, showing that zNfL changes reflect focal inflammatory activity and are associated with commonly used measures of treatment response, in line with findings from clinical trials and large observational cohorts.^{4,5,13–15,26,27}

In our analyses, persistently elevated zNfL levels after 1 year were consistently associated with disease activity at year 2, independent of conventional clinical and radiologic measures and across treatment categories. Associations were more pronounced in treatment-naïve patients, possibly reflecting higher inflammatory activity and lower clinical heterogeneity compared with previously treated cohorts. Importantly, the association remained significant in previously treated patients, supporting the broader clinical relevance of NfL dynamics across treatment contexts. A reduction of ≥ 0.5 points in zNfL after 1 year was generally associated with lower odds of subsequent disease activity, although effect estimates attenuated after adjustment for treatment category and overall discriminative performance was modest (area under the curve [AUC] 0.63). These findings should therefore be interpreted as associative rather than causal.²⁸

In line with prior literature, a single zNfL measurement at year 1 above 1.5 was also associated with subsequent disease activity, suggesting that cross-sectional assessment may provide clinically relevant information. However, longitudinal changes showed more consistent associations across analyses—including in patients otherwise considered stable—and may better capture treatment-related biological effects over time.

An important observation from this study is the added value of zNfL changes in patients considered stable or good responders. In individuals achieving NEDA-3 or with minimal evidence of disease activity, Δ zNfL identified patients at higher risk of subsequent disease activity, a clinical context in which prognostic assessment may be uncertain.²⁹ Current MS monitoring methods may not capture the full spectrum of disease activity, and our results suggest that integrating zNfL dynamics with existing scoring systems may refine assessment of ongoing activity and treatment response across different baseline risk profiles.

Figure 4 Association of Year-1 zNfL Changes With Evidence of Disease Activity When Combined With Treatment Response Scoring Systems

Forest plots show multivariable logistic regression models combining dichotomized Rio (RS), modified Rio (mRS), and MAGNIMS scores (low vs high) with $\Delta zNfL$ at year 1 and the association with EDA-Y2. Odds ratios (ORs) with 95% confidence intervals (CI) are displayed. (A–C) TRSS combined with $\Delta zNfL$ using the absence of reduction cut-off ($\Delta zNfL \geq 0$) for RS (A), mRS (B), and MAGNIMS (C). Numbers of patients in each category were Low score + zNfL reduction: RS (n = 99), mRS (n = 107), MAGNIMS (n = 111). Low score + no zNfL reduction: RS (n = 47), mRS (n = 55), MAGNIMS (n = 58). High score + zNfL reduction: RS (n = 18), mRS (n = 14), MAGNIMS (n = 14). High score + no zNfL reduction: RS (n = 15), mRS (n = 7), MAGNIMS (n = 9). (D–F) TRSS combined with $\Delta zNfL$ using the ≥ 0.5 -point reduction threshold for RS (D), mRS (E), and MAGNIMS (F). Numbers of patients in each category were Low score + zNfL reduction: RS (n = 66), mRS (n = 74), MAGNIMS (n = 76). Low score + no zNfL reduction: RS (n = 80), mRS (n = 88), MAGNIMS (n = 93). High score + zNfL reduction: RS (n = 15), mRS (n = 12), MAGNIMS (n = 12). High score + no zNfL reduction: RS (n = 18), mRS (n = 9), MAGNIMS (n = 11). $\Delta zNfL$ = changes in neurofilament light chain z-scores at year 1; EDA = evidence of disease activity; TRSS = treatment response scoring systems; Y2 = year 2. ***: odds ratio not estimable.

From a clinical perspective, these findings are applicable at the individual patient level because they rely on intraindividual longitudinal changes rather than absolute thresholds. This approach may overcome limitations of single time-point cut-offs, which vary across populations and clinical contexts. Age and BMI-adjusted zNfL values can be readily derived using available reference tools, facilitating implementation in routine practice.⁹

$\Delta zNfL$ at year 1 was not associated with disease activity at year 3, suggesting that early changes in NfL may not reliably inform longer-term disease control in isolation. Prior studies have shown that patients treated with moderate efficacy DMTs may show an initial decline in NfL levels but later on may fail to maintain an adequate disease control, whereas high-efficacy therapies are associated with sustained normalization of NfL

toward healthy reference ranges.⁹ These observations support the need for repeated longitudinal reassessments of NfL, analogous to the use of rebaseline and repeated MRI scans in long-term treatment monitoring.

As expected, platform injectable therapies were associated with substantially higher odds of subsequent disease activity compared with monoclonal antibodies, consistent with their known differences in efficacy. This observation likely reflects treatment era and prescribing patterns rather than a direct biological effect but also underscores the importance of early optimization of therapeutic strategies.

Several limitations should be acknowledged. The cohort consisted of patients with RMS initiating a new DMT with available serum samples at both baseline and year 1; exclusions were

mainly because of missing samples, early treatment discontinuation, or incomplete follow-up. Although extensive sensitivity analyses—including multivariable adjustment, inverse-probability-of-attrition weighting, and bootstrap resampling—supported the robustness of the findings, residual selection bias cannot be fully excluded.

Incomplete MRI availability at certain time points may have reduced statistical power for specific analyses. MRI scans were accepted within a ± 180 -day window around baseline and year 1 to maximize inclusion, consistent with prior studies^{15,30}; however, this approach may have introduced temporal variability in the assessment of MRI activity. By contrast, the interval between MRI acquisition and sNfL sampling—more relevant for biomarker–MRI comparisons—was substantially shorter in practice, particularly at year 1, supporting adequate temporal alignment. The absence of systematic spinal cord MRI may have led to under-ascertainment of inflammatory activity.

sNfL levels may be influenced by non-neurologic factors (e.g., infections, renal function, BMI), although the use of longitudinal intraindividual changes may partially mitigate this limitation.

Discrimination of both $\Delta zNfL$ and year 1 $zNfL$ for EDA was modest (AUC ~ 0.63 – 0.64), which is expected for a single biomarker evaluated against a multifactorial clinical–radiologic outcome and supports a complementary rather than stand-alone role for $zNfL$ dynamics. Selected operational cut-offs were derived from the plateau of maximal Youden index values and showed consistent associations across adjusted and sensitivity analyses.

Finally, the inclusion period ended in 2021 and may not fully reflect current treatment algorithms.

In summary, observed changes in $zNfL$ after 1 year of treatment were associated with subsequent disease activity and provided information complementary to conventional clinical and radiologic measures. $\Delta zNfL$ may represent a useful adjunct for treatment monitoring when interpreted within a multimodal assessment framework.

Acknowledgment

We thank Marta Rodríguez Barranco for her assistance in data acquisition and to Prof. Jens Kuhle and Pascal Benkert for performing the z-score calculations based on the provided concentration values. ChatGPT (OpenAI, GPT-5.5) was used to improve the clarity and readability of the manuscript. The scientific content, interpretation, and conclusions were written entirely by the authors. All generated text was reviewed and approved by the authors prior to submission.

Author Contributions

V. Fernández: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or

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content; study concept or design. G. Arrambide: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data.

Study Funding

This study has been funded by Instituto de Salud Carlos III (ISCIII) through the project “PI19/01590” and cofunded by the European Union.

Disclosure

V. Fernández was anECTRIMS Clinical Training Fellowship awardee (2022–2023) and has received travel support from Novartis, UCB and Merck; A. Pappolla has received travel funding from Roche and speaking honoraria from Novartis and completed anECTRIMS Clinical Training Fellowship (2021–2022), an MSIF-ARSEP Fellowship (2023–2024), and received a Research Training Programme award from the European Charcot Foundation (2025); P. Carbonell-Mirabent’s salary is supported by a grant from Biogen to Fundació Privada Cemcat for statistical analysis; M. Castillo reports no disclosures relevant to the manuscript; L. Gutiérrez reports no disclosures relevant to the manuscript; N. Fissolo reports no disclosures relevant to the manuscript; O. León Plaza has received travel support from Merck, Novartis, and Neuraxpharm; N. Braga has received a fellowship grant fromECTRIMS and speaking honoraria and travel support from Merck, Janssen, and AstraZeneca; H. Ariño has received a grant from Instituto de Salud Carlos III, Spain (JR22/00072); C. Auger reports no disclosures relevant to the manuscript; L. Bollo was supported by a stipend from the NMSS/AAN John Dystel Prize (2022) and has received a Río Hortega grant from Instituto de Salud Carlos III (CM24/00255); J. Castilló reports no disclosures relevant to the manuscript; R. Carvajal has received funding from Instituto de Salud Carlos III (FI/2400299), a European Charcot Foundation Research Training Programme (2023), a Vall d’Hebron Institut de Recerca grant (2023), and anECTRIMS Fellowship (2021–2022), and has received speaking honoraria from Roche, Novartis, Biogen, Merck, and Sanofi; A. Cobo-Calvo has received a grant from Instituto de Salud Carlos III, Spain (JR19/00007); C. Espejo reports no disclosures relevant to the manuscript; I. Galán reports no disclosures relevant to the manuscript; C. Guío-Sánchez was anECTRIMS Clinical Fellowship awardee (2022–2023) and has received travel support from Sanofi-Genzyme, Biogen, and Merck; D. La Puma reports no disclosures relevant to the manuscript; L. Midaglia reports no disclosures relevant to the manuscript; N. Mongay-Ochoa has received anECTRIMS-MAGNIMS Research Fellowship (2024); C. Nos reports no disclosures relevant to the manuscript; S. Otero-Romero has received speaking and consulting honoraria from Genzyme, Biogen, Novartis, Roche, Excemed, and MSD, and research support from Novartis; B. Rodríguez-Acevedo has received speaking honoraria from Merck and consulting honoraria from Novartis; J. Sastre-Garriga serves as co-Editor for Europe of Multiple Sclerosis Journal and Editor-in-Chief of Revista de

Neurología, has received research support from Fondo de Investigaciones Sanitarias (19/950, 22/750), and has served as a consultant or speaker for BMS, Roche, Sanofi, Janssen, and Merck; P. Tagliani received support as anECTRIMS Clinical Fellowship awardee (2019–2020); C. Tur is funded by a Miguel Servet contract (CP23/00117, Instituto de Salud Carlos III) and has received multiple research grants and fellowships, and honoraria from Roche, Novartis, Merck, Immunic Therapeutics, and Bristol Myers Squibb, and serves on editorial boards and steering committees; A. Vidal-Jordana has received support from Instituto de Salud Carlos III (JR16/00024; PI17/02162; PI22/01589) and has received consulting and speaking honoraria from Roche, Novartis, Merck, and Sanofi; A. Vilaseca reports no disclosures relevant to the manuscript; A. Villaceros-Álvarez has received a grant from Instituto de Salud Carlos III (FI21/00282); A. Zabalza has received a Río Hortega grant (CM22/00237), travel support from Biogen, Merck, and Novartis, speaking honoraria from Eisai, and a study grant from Novartis; À. Rovira serves on advisory boards for multiple companies, has received speaker honoraria, is co-founder of TensorMedical, and receives research support from Instituto de Salud Carlos III (PI19/00950); M. Tintoré has received consulting fees, speaking honoraria, and research support from multiple pharmaceutical companies and has served on data safety monitoring boards; M. Comabella has received consulting and speaking honoraria from Bayer, Merck, Biogen, Teva, Sanofi, and Novartis; J. Río has received speaking honoraria and advisory board compensation from Biogen, Genzyme, Janssen, Merck, Novartis, Teva, Roche, and Sanofi; X. Montalban’s institution has received compensation for consulting, advisory boards, and clinical trial participation from multiple pharmaceutical companies and organizations; G. Arrambide has received consulting fees, speaking honoraria, and travel support from Roche, Bristol Myers Squibb, UCB, Novartis,ECTRIMS, and EAN, and serves on multiple editorial boards and scientific committees. Go to [Neurology.org/N](https://www.neurology.org/N) for full disclosures.

Publication History

Received by *Neurology*® February 17, 2026. Accepted in final form June 4, 2026. Submitted and externally peer reviewed. The handling editor was Deputy Editor Olga Ciccarelli, MD, PhD, FRCP.

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